

Akari Imai

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EDUCATION

Duke University

Durham, NC

Double Major in Biomedical Engineering and Mechanical Engineering

Expected May 2029

- Relevant Coursework: Algorithms & Data Structures, Engineering Design & Technical Communication, Matrices & Vectors

Phillips Academy Andover

Andover, MA

High School Diploma; GPA 5.5/6.0

Aug. 2022 – Jun. 2025

EXPERIENCE

Technical Project Manager Intern

Jul. 2026 – Present

Stēle Inc. – In-Ear EEG Wearable Startup

Culver City, CA

- Coordinate 8 engineers across ML, hardware, and app teams via weekly reports and standups; onboarded 5 engineers
- Track every project's owner, status, and deadline; contributed to an invention under a provisional patent application

Intern

Jun. 2023

Effissimo Capital Management Pte Ltd

Singapore

- Designed, built, and back-tested a machine-learning model forecasting the Japanese stock market from multiple valuation metrics; strategy returned 23.2% annualized, 14.4 percentage points above the TOPIX 500 benchmark

RESEARCH

Visiting Researcher

Jan. 2026 – Jul. 2026

Yokohama National University, Institute of Multidisciplinary Sciences

Yokohama, Japan

- First-authored an accepted Work-in-Progress paper for IEEE ISIE 2026 on the surgeon-centered muscular benefits of automated penetration detection in a haptic spinal drill; presented as the conference's youngest participant
- Showed that automated penetration detection cut posterior deltoid and extensor carpi radialis brevis muscular demand by 11.96 and 3.96%MVC, respectively, by recording surface electromyography (sEMG) during haptic spinal drill operation

Independent Researcher

Dec. 2024 – May 2025

Phillips Academy Andover

Andover, MA

- Found dose-dependent cell death and a 28.6% increase in DNA degradation across concentrations by exposing 4T1 mouse mammary cells to cycloxyaprid, a novel neonicotinoid insecticide, and assaying cell viability and DNA laddering
- DNA laddering implied apoptotic and necrotic cytotoxicity; reported findings in a 20+ page research paper

Student Researcher

Jun. 2024 – Jul. 2024

Keio University, Frontier Research & Education Collaborative Square

Kanagawa, Japan

- Raised penetration detection accuracy to 100% at all three drilling angles, up from 42.9% at 45° and 14.3% at 90°, by adding gravity compensation with accelerometer-based slope estimation to a haptic drill robot designed for spinal surgery
- First-authored the resulting manuscript, currently being submitted to the IEEE Journal of Industry Applications

Independent Researcher

Nov. 2023 – Jun. 2025

Supervised by Mackenzie Simper, WashU MD student

Singapore

- Retrained a published random-forest binding-affinity model on ChEMBL data for the human androgen receptor and androgen-binding protein, then screened 33 literature-curated pesticides against 1,440 background compounds
- Predicted above-average receptor binding for acetamiprid and three banned organochlorines; wrote a sole-author paper

LEADERSHIP & VOLUNTEERING

Co-President & Head of Hardware | Phillips Academy Robotics (FRC & FTC)

- Taught 20+ beginners CAD and machining; recruited 7 female students; team won Think, Judges', and Quality Awards

Co-President | Andover Japanese Connection

Dec. 2023 – May 2025

- Revived an inactive club, growing it to 30+ members with cultural events and lessons; raised \$400+ for disaster relief

Co-Founder & Coach | FIRST LEGO League Robotics, Andover & Brookline, MA

Aug. 2022 – Dec. 2024

- Co-founded a program and curriculum for 30+ elementary and middle schoolers; coached a team to state qualification

Volunteer English Tutor | KIDS DOOR, Japan

Jan. 2026 – Apr. 2026

- Tutored ~10 underprivileged children one-to-one in Japanese, teaching English and structuring lessons catered to each student

SKILLS

Technical: Python, Java, C++, Fusion 360, Onshape, Excel, Cell Culture, DNA Laddering, Surface Electromyography (sEMG)

Spoken Languages: English, Japanese, Chinese